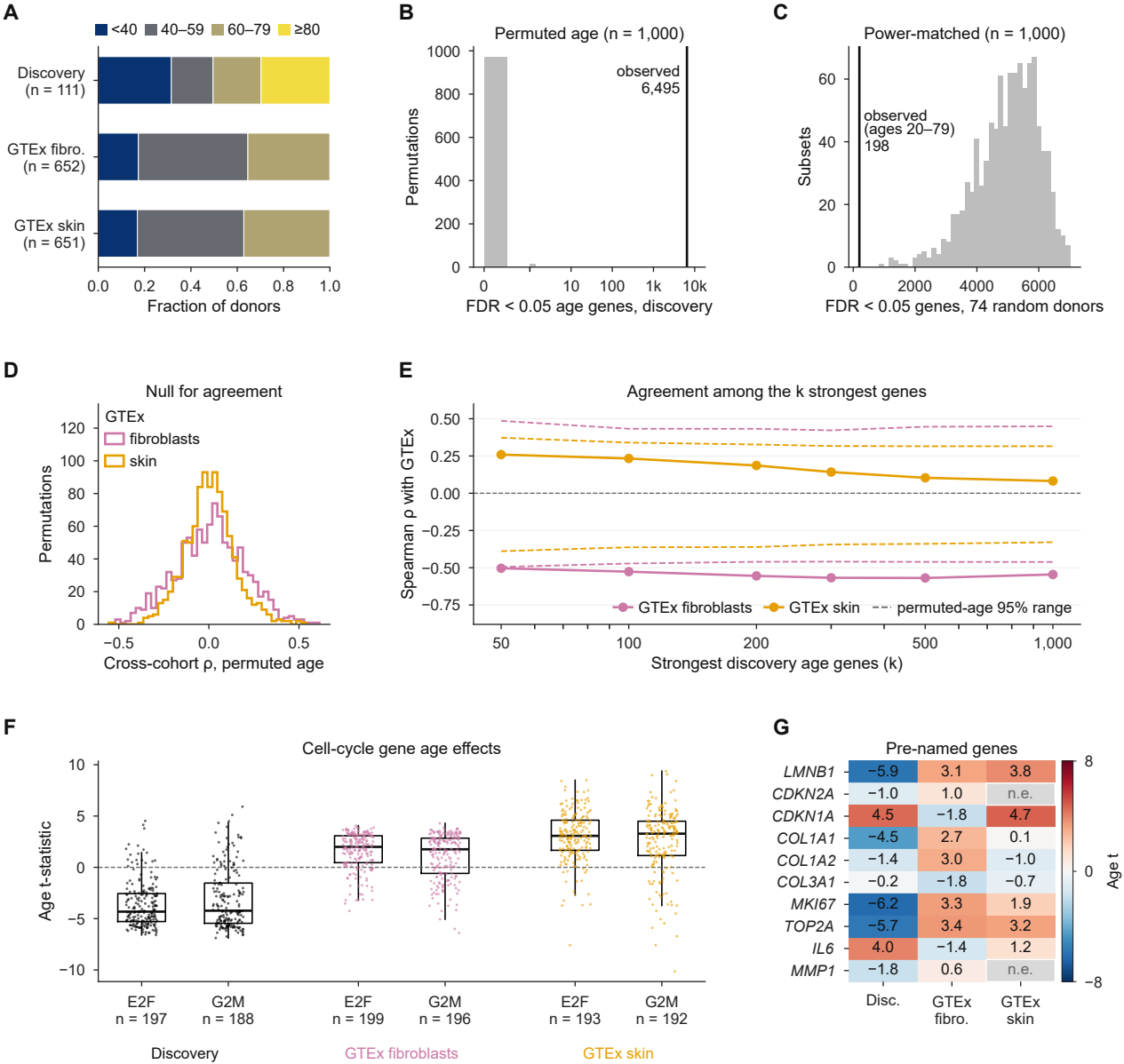

A
<40
40–59
60–79
≥80
Discovery (n = 111)
GTEx fibro. (n = 652)
GTEx skin (n = 651)
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of donors
B
Permuted age (n = 1,000)
observed 6,495
Permutations
1000
800
600
400
200
0
0
10
100
1k
10k
FDR < 0.05 age genes, discovery
C
Power-matched (n = 1,000)
observed (ages 20–79) 198
Subsets
60
40
20
0
0
2000
4000
6000
FDR < 0.05 genes, 74 random donors
D
Null for agreement
GTEx
fibroblasts
skin
Permutations
120
100
80
60
40
20
0
−0.5
0.0
0.5
Cross-cohort ρ, permuted age
E
Agreement among the k strongest genes
Spearman ρ with GTEx
0.50
0.25
0.00
−0.25
−0.50
−0.75
GTEx fibroblasts
GTEx skin
permuted-age 95% range
50
100
200
500
1,000
Strongest discovery age genes (k)
F
Cell-cycle gene age effects
Age t-statistic
10
5
0
−5
−10
E2F n = 197
G2M n = 188
E2F n = 199
G2M n = 196
E2F n = 193
G2M n = 192
Discovery
GTEx fibroblasts
GTEx skin
G
Pre-named genes
LMNB1 −5.9 3.1 3.8
CDKN2A −1.0 1.0 n.e.
CDKN1A 4.5 −1.8 4.7
COL1A1 −4.5 2.7 0.1
COL1A2 −1.4 3.0 −1.0
COL3A1 −0.2 −1.8 −0.7
MKI67 −6.2 3.3 1.9
TOP2A −5.7 3.4 3.2
IL6 4.0 −1.4 1.2
MMP1 −1.8 0.6 n.e.
Disc.
GTEx fibro.
GTEx skin
8
0
−8
Age t